1.3 Coordinate Geometry

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

Coordinate geometry translates geometric objects — lines, circles, and parametric curves — into algebraic equations in the Cartesian (x, y) plane. The philosophy is Descartes': every geometric statement becomes algebraic, every algebraic statement becomes a picture. At A-level this topic blends synthetic-geometry reasoning (angles, perpendicularity, radii) with algebraic techniques (completing the square, discriminants, substitution) and sets the stage for calculus applications later in the course.

1. Straight Lines

A line in the plane is uniquely determined by either two distinct points or one point together with a direction (gradient). The gradient measures *rise over run*: the vertical change per unit horizontal change.

Gradient. For two distinct points $A(x_1, y_1)$ and $B(x_2, y_2)$ with $x_1 \neq x_2$, the gradient of the line AB is \[m = \frac{y_2 - y_1}{x_2 - x_1}. \] If $x_1 = x_2$ the line is vertical and its gradient is undefined; if $y_1 = y_2$ the line is horizontal with gradient 0.

Three Standard Forms of a Line

Form	Equation	Best used when...
Slope–intercept	$y = mx + c$	Gradient m and y-intercept c are known; also best for sketching.
Point–slope	$y - y_1 = m(x - x_1)$	One point (x_1, y_1) and the gradient m are known.
General (implicit)	$ax + by + c = 0$	Required final form in some OCR questions; handles vertical lines ($b = 0$).

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Two-point	$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$	Two points are given; equivalent to point–slope after computing m.

Exam tip — which form to use. OCR mark schemes usually accept any correct form unless a specific form is demanded. However, a question ending "...giving your answer in the form $ax+by+c=0$ where a, b, c are integers" means the implicit form with integer coefficients — not $y = 2.5x - 1.5$. Multiply through to clear fractions and move everything to the left-hand side.

Parallel and Perpendicular Lines

Two non-vertical lines are **parallel** iff they have equal gradients: $m_1 = m_2$. They are **perpendicular** iff the product of gradients is -1.

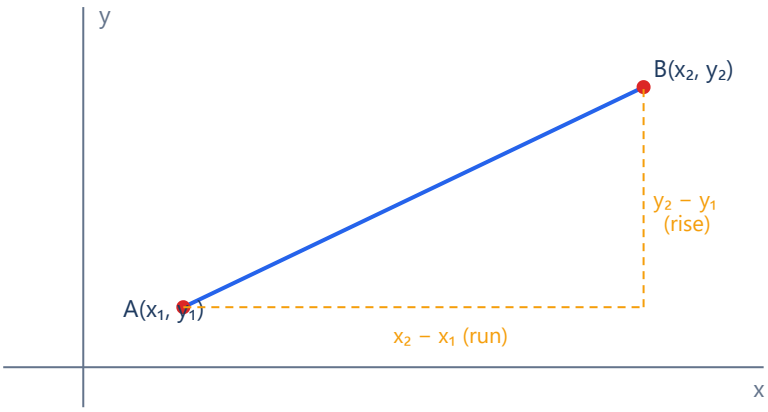
Perpendicularity condition. Two non-vertical, non-horizontal lines with gradients m_1 and m_2 are perpendicular if and only if \[m_1 \cdot m_2 = -1, \qquad \text{equivalently} \qquad m_2 = -\frac{1}{m_1}. \] A horizontal line ($m = 0$) is perpendicular to a vertical line (gradient undefined); this is the limiting case not captured by the product rule.

Distance Between Two Points and Midpoint

Given $A(x_1, y_1)$ and $B(x_2, y_2)$:

$$|AB| = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}, \qquad M_{AB} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right).$$

The distance formula is Pythagoras applied to the right triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$. The midpoint is simply the coordinate-wise average.



Worked Example 1 — Line through two points

Find the equation of the line through $A(-1, 4)$ and $B(3, -2)$ in the form $ax + by + c = 0$ with integer a, b, c. State the midpoint and the distance $|AB|$.

```
Gradient  m = (-2 - 4) / (3 - (-1))  =  -6 / 4  =  -3/2.
Point-slope using A(-1, 4):
   y - 4 = -3/2 · (x - (-1))
   y - 4 = -3/2 · (x + 1)
   2(y - 4) = -3(x + 1)
   2y - 8  = -3x - 3
   3x + 2y - 5 = 0.           // final form

Midpoint  M = ((-1 + 3)/2, (4 + (-2))/2)  =  (1, 1).
Distance  |AB| = √[(3 - (-1))² + (-2 - 4)²]
               = √[16 + 36]  =  √52  =  2√13.
Check M lies on the line:  3(1) + 2(1) - 5 = 0 ✓.
```

Intersection of Two Lines

Two lines in the plane either (i) intersect at one point — non-parallel; (ii) are identical — infinitely many intersection points; (iii) are parallel and distinct — no intersection. To find the intersection, solve the system simultaneously (substitution or elimination).

Worked Example 2 — Perpendicular through a point

Find the equation of the line through $(1, 2)$ perpendicular to $y = 3x - 1$.

```
Gradient of given line:  m₁ = 3.
Perpendicular gradient:  m₂ = -1/m₁ = -1/3.
Point-slope at (1, 2):
   y - 2 = -1/3 · (x - 1)
   3(y - 2) = -(x - 1)
   3y - 6   = -x + 1
   x + 3y - 7 = 0.          // or y = -x/3 + 7/3
Check perpendicularity:  3 · (-1/3) = -1 ✓.
Check point:  1 + 3·2 - 7 = 0 ✓.
```

2. Circles — Standard and General Forms

A circle is the locus of points equidistant from a fixed centre. Translating this definition using the distance formula gives the equation of a circle.

Circle — standard (centre–radius) form. The circle with centre (a, b) and radius $r > 0$ has equation \[(x - a)^2 + (y - b)^2 = r^2. \] A point (x_0, y_0) lies on the circle iff the equation is satisfied; inside iff LHS $< r^2$; outside iff LHS $> r^2$.

Circle — general form. Expanding the standard form gives \[x^2 + y^2 + 2gx + 2fy + c = 0, \] with centre $(-g, -f)$ and radius $r = \sqrt{g^2 + f^2 - c}$ (a real circle requires $g^2 + f^2 - c > 0$; equality gives a point, and a negative value gives no real locus).

Comparison of Forms

Feature	Standard form $(x - a)^2 + (y - b)^2 = r^2$	General form $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre	(a, b) — read directly	$(-g, -f)$ — half the coefficient of x, negated
Radius	r — square root of RHS	$r = \sqrt{g^2 + f^2 - c}$
Convenient for...	Sketching, geometric reasoning	Substituting a line equation; line–circle intersection
How to convert	Expand the brackets	Complete the square in x and y separately
Validity check	$r^2 > 0$	$g^2 + f^2 - c > 0$

Exam tip — completing the square for a circle. OCR questions frequently hand you a circle in general form (e.g. $x^2 + y^2 - 4x + 6y - 3 = 0$) and ask for centre and radius. Complete the square in x and y *separately*: group $x^2 - 4x$ and add/subtract 4; group $y^2 + 6y$ and add/subtract 9; move the constants. Many students forget to subtract what they added and end up with the wrong r^2.

Worked Example 3 — General to standard form

Find the centre and radius of the circle $x^2 + y^2 - 4x + 6y - 3 = 0$.

```
Group:          (x² − 4x) + (y² + 6y) − 3 = 0.
Complete the square in x:   x² − 4x = (x − 2)² − 4.
Complete the square in y:   y² + 6y = (y + 3)² − 9.
Substitute:   (x − 2)² − 4 + (y + 3)² − 9 − 3 = 0
              (x − 2)² + (y + 3)² = 16.
∴ centre (2, −3), radius r = 4.

// Sanity check using g, f, c directly:
Compare with x² + y² + 2gx + 2fy + c = 0:  2g = −4, 2f = 6, c = −3.
g = −2, f = 3, so centre = (−g, −f) = (2, −3).  ✓
r² = g² + f² − c = 4 + 9 − (−3) = 16, so r = 4.  ✓
```

Point Lying on / Inside / Outside a Circle

Given a circle with centre $C(a, b)$ and radius r, and a point $P(x_0, y_0)$, compute the distance $d = |CP|$:

- $d = r$ — P lies on the circle;
- $d < r$ — P lies inside the circle;
- $d > r$ — P lies outside the circle.

In practice you compare d^2 with r^2 to avoid square roots.

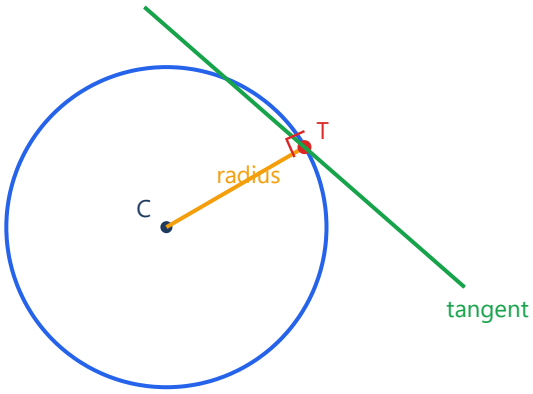
3. Circle Theorems Relevant to Coordinate Geometry

Three classical Euclidean theorems translate cleanly into coordinate-geometry tools. They are not proved here in detail (see GCSE circle theorems) but stated in the forms you will deploy.

Theorem A — angle in a semicircle. If AB is a diameter of a circle and P is any other point on the circle, then $\angle APB = 90^\circ$. Conversely, if $\angle APB = 90^\circ$ for points A, B, P, then P lies on the circle with diameter AB.

Theorem B — radius perpendicular to chord. The perpendicular from the centre of a circle to a chord bisects the chord. Conversely, the perpendicular bisector of any chord passes through the centre.

Theorem C — tangent perpendicular to radius. The tangent to a circle at a point T is perpendicular to the radius drawn to T. Thus if the radius CT has gradient m, the tangent at T has gradient $-1/m$ (when neither is vertical or horizontal).



Tangent at T is perpendicular to the radius CT.

Applications

- **Equation of tangent at a given point**: gradient of tangent = $-1/$(gradient of radius from centre to point); use point–slope form.
- **Centre via perpendicular bisectors**: for three points on a circle, the centre is the intersection of any two perpendicular bisectors.
- **Right-angle detection**: if three points satisfy Pythagoras' theorem, the right-angle vertex lies on the circle whose diameter is the hypotenuse.

Exam tip — tangent at a point. If a question asks for the equation of the tangent to a circle at a specific point T on the circle, do *not* substitute $y = mx + c$ into the circle equation and set the discriminant to zero. Use the radius-perpendicular-to-tangent theorem: compute the gradient of CT, take the negative reciprocal, and apply point–slope at T. This is faster and less error-prone than the discriminant approach, which is reserved for tangents from an *external* point.

Worked Example 4 — Tangent at a point on a circle

Find the equation of the tangent at $T(5, 6)$ to the circle $(x-2)^2 + (y-2)^2 = 25$.

```
1.  Verify T lies on the circle:  (5-2)² + (6-2)² = 9 + 16 = 25 ✓.
2.  Centre C = (2, 2). Gradient of radius CT:
3.     m(CT) = (6 - 2)/(5 - 2) = 4/3.
4.  Gradient of tangent at T = -3/4 (negative reciprocal).
5.  Tangent equation, point-slope at (5, 6):
```

```
    y − 6 = −3/4 · (x − 5)
    4(y − 6) = −3(x − 5)
    4y − 24  = −3x + 15
    3x + 4y − 39 = 0.
```

4. Intersection of a Line and a Circle

To find the intersection(s) of a line L and a circle C: solve the line for y (or x) in terms of the other variable and substitute into the circle equation. The result is a quadratic in one variable, whose discriminant Δ classifies the geometry.

Sign of Δ	Number of intersections	Geometric meaning
$\Delta > 0$	Two distinct points	Line is a *secant* (cuts the circle)
$\Delta = 0$	One (repeated) point	Line is a *tangent* to the circle
$\Delta < 0$	None	Line misses the circle entirely

Exam tip — tangent from an external point. When the line passes through an *external* point (p, q) and is tangent to the circle, parametrise it as $y - q = m(x - p)$, substitute into the circle equation, and set the discriminant equal to zero. You'll get a quadratic in m — two real solutions giving the two tangent gradients from the external point. Do not forget that the vertical line $x = p$ may also be a tangent; check separately if the algebra hides this case.

Worked Example 5 — Tangents from an external point

Find the equations of both tangents from the external point $P(7, 1)$ to the circle $x^2 + y^2 = 25$.

```
Circle: x² + y² = 25, centre (0,0), radius 5.
|OP|² = 49 + 1 = 50 > 25, so P is external. ✓
Line through P with gradient m:   y − 1 = m(x − 7), i.e. y = mx − 7m + 1.
Substitute into x² + y² = 25:
   x² + (mx − 7m + 1)² = 25.
```

```
Let k = 1 − 7m.  Then y = mx + k.
   x² + m²x² + 2mkx + k² = 25
   (1 + m²)x² + 2mk x + (k² − 25) = 0.
Tangent ⇒ discriminant = 0:
   (2mk)² − 4(1 + m²)(k² − 25) = 0
   4m²k² − 4(k² − 25 + m²k² − 25m²) = 0
   4m²k² − 4k² + 100 − 4m²k² + 100m² = 0
   −4k² + 100 + 100m² = 0
   k² = 25(1 + m²).
Substitute k = 1 − 7m:
   (1 − 7m)² = 25(1 + m²)
   1 − 14m + 49m² = 25 + 25m²
   24m² − 14m − 24 = 0
   12m² − 7m − 12 = 0.
Solve: m = [7 ± √(49 + 576)] / 24  =  [7 ± 25]/24.
⇒ m = 32/24 = 4/3   or   m = −18/24 = −3/4.
Tangent equations:
  y − 1 =  4/3 (x − 7)   ⇒   4x − 3y − 25 = 0.
  y − 1 = −3/4 (x − 7)   ⇒   3x + 4y − 25 = 0.
Check product of gradients: (4/3)·(−3/4) = −1, so the two tangents are perpe
```

Worked Example 6 — Circle through three points

Find the equation of the circle through $A(1,1)$, $B(5,1)$, and $C(1,5)$.

```
Strategy A — general form.  Substitute each point into
    x² + y² + 2gx + 2fy + c = 0.
   A(1,1):  1 + 1 + 2g + 2f + c = 0   ⇒   2g + 2f + c = −2.
   B(5,1): 25 + 1 + 10g + 2f + c = 0  ⇒  10g + 2f + c = −26.
   C(1,5):  1 + 25 + 2g + 10f + c = 0 ⇒   2g + 10f + c = −26.
Subtract (2) from (3):  8g = −24  ⇒  g = −3.
Subtract (2) from (4):  8f = −24  ⇒  f = −3.
From (2):  −6 − 6 + c = −2  ⇒  c = 10.
∴ x² + y² − 6x − 6y + 10 = 0,
 centre (3, 3), r² = 9 + 9 − 10 = 8, r = 2√2.

// Strategy B — perpendicular bisectors (for geometric insight).
AB is horizontal, midpoint (3, 1), so its perpendicular bisector is x = 3.
```

```
AC is vertical, midpoint (1, 3),  so its perpendicular bisector is y = 3.
Centre = (3, 3), agreeing with Strategy A.
Radius = distance from (3,3) to A(1,1) = √(4 + 4) = 2√2.  ✓
```

5. Parametric Equations

Not every curve in the plane is conveniently described by a single equation $y = f(x)$ or $F(x, y) = 0$. *Parametric equations* describe a curve by letting both x and y depend on a third variable, the *parameter*, often called t or θ.

Parametric equations. A curve given by \[x = f(t), \quad y = g(t), \qquad t \in I, \] for some interval (or union of intervals) $I \subseteq \mathbb{R}$. As t varies over I, the point (x, y) traces out the curve.

Parameter. The independent variable t (or θ, s, λ, etc.) indexing the points of a parametric curve. The parameter often has a physical or geometric meaning — time in mechanics; angle for circles and ellipses; a non-geometric algebraic label otherwise.

Standard Parametrisations

Curve	Parametric form	Cartesian equation	Elimination technique
Parabola	$x = t^2,\ y = 2t$	$y^2 = 4x$	From $y = 2t$, $t = y/2$; substitute
Circle, radius a	$x = a\cos t,\ y = a\sin t$	$x^2 + y^2 = a^2$	Pythagorean identity $\sin^2 t + \cos^2 t = 1$
Ellipse	$x = a\cos t,\ y = b\sin t$	$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$	$(x/a)^2 + (y/b)^2 = 1$
Rectangular hyperbola	$x = t,\ y = c^2/t$	$xy = c^2$	Multiply x and y

Line	$x = p + \lambda u,\ y = q + \lambda v$	$v(x-p) - u(y-q) = 0$	Solve one for λ, substitute

Conversion Techniques — Parametric to Cartesian

Parametric feature	Elimination method
Polynomial in t, one equation easily invertible	Solve for t in one equation, substitute into the other
Both x and y are trigonometric (e.g. $\cos t, \sin t$)	Use $\sin^2 t + \cos^2 t = 1$, or $1 + \tan^2 t = \sec^2 t$
Exponential parameters (e.g. e^t, e^{-t})	Multiply, or use $\cosh^2 - \sinh^2 = 1$
Involves reciprocals t and $1/t$	Add or multiply to produce a simple symmetric function

Exam tip — range of the curve. Eliminating the parameter gives an equation $F(x, y) = 0$ that the points of the curve must satisfy — but the Cartesian equation often describes a *larger* set than the original parametric curve. For example, $x = t^2,\ y = t$ with $t \in \mathbb{R}$ yields $x = y^2$ — the full sideways parabola. But $x = t^2,\ y = t,\ t \geq 0$ traces only the upper half ($y \geq 0$). Always state any restrictions on x or y inherited from the parameter's range.

Worked Example 7 — Trigonometric parametrisation

Show that $x = 4\cos\theta,\ y = 4\sin\theta$ traces a circle; give its Cartesian equation and state its centre and radius.

```
1.   x² = 16 cos²θ,     y² = 16 sin²θ.
2.   x² + y² = 16(cos²θ + sin²θ) = 16 · 1 = 16.
3.   ∴ x² + y² = 16  –  circle, centre (0, 0), radius 4.

// As θ runs over [0, 2π), (x, y) traces the whole circle exactly once.
// θ = 0 → (4, 0); θ = π/2 → (0, 4); θ = π → (−4, 0); θ = 3π/2 → (0, −4).
```

Worked Example 8 — Parabolic parametrisation

A curve is given by $x = t^2,\ y = 2t$. (a) Find its Cartesian equation. (b) Find the points where it meets the line $y = x - 3$.

```
// (a) Eliminate t.
1.  From y = 2t:  t = y/2.
2.  Substitute into x = t²:  x = (y/2)² = y²/4.
3.  ∴ y² = 4x  –  a parabola, vertex at origin, axis along the x-axis.

// (b) Intersection with y = x – 3.
4.  Substitute x = y + 3 into y² = 4x:
5.     y² = 4(y + 3)
6.     y² – 4y – 12 = 0
7.     (y – 6)(y + 2) = 0
8.  ⇒ y = 6, x = 9    or    y = –2, x = 1.
9.  Intersection points:  (9, 6) and (1, –2).
```

6. Differentiation of Parametric Curves

When x and y are both functions of t, the gradient $\mathrm{d}y/\mathrm{d}x$ of the curve can be computed without first eliminating the parameter.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}, \qquad \text{provided } \frac{\mathrm{d}x}{\mathrm{d}t} \neq 0.$$

This is the chain rule in disguise: $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y}{\mathrm{d}t} \cdot \dfrac{\mathrm{d}t}{\mathrm{d}x}$. When $\mathrm{d}x/\mathrm{d}t = 0$ the tangent is vertical — the formula breaks down, which it should, because the gradient is undefined.

Exam tip — parametric $\mathrm{d}y/\mathrm{d}x$. Always write out both $\mathrm{d}y/\mathrm{d}t$ and $\mathrm{d}x/\mathrm{d}t$ separately on the page, then divide. OCR marks the derivatives and the division as separate steps. Also: leave your answer in terms of t (not x or y) unless asked to convert — converting midway adds unnecessary algebra and is a common source of errors.

Worked Derivatives

Curve	$\mathrm{d}x/\mathrm{d}t$	$\mathrm{d}y/\mathrm{d}t$	$\mathrm{d}y/\mathrm{d}x$
Parabola: $x = t^2,\ y = 2t$	$2t$	2	$1/t$ (equivalently $2/y$)
Circle: $x = a\cos t,\ y = a\sin t$	$-a\sin t$	$a\cos t$	$-\cot t = -x/y$
Ellipse: $x = a\cos t,\ y = b\sin t$	$-a\sin t$	$b\cos t$	$-\dfrac{b}{a}\cot t$
Rect. hyperbola: $x = t,\ y = c^2/t$	1	$-c^2/t^2$	$-c^2/t^2 = -y/x$

Worked Example 9 — Tangent to a parametric curve

Find the equation of the tangent to the curve $x = t^2,\ y = 2t$ at the point where $t = 3$.

```
1.  Point at t = 3:  (x, y) = (9, 6).
2.  dx/dt = 2t,   dy/dt = 2.
3.  dy/dx = 2/(2t) = 1/t.
4.  At t = 3:   dy/dx = 1/3.
5.  Tangent:   y − 6 = (1/3)(x − 9)
6.             3y − 18 = x − 9
7.             x − 3y + 9 = 0.
8.  Check on Cartesian form y² = 4x: implicit differentiation
        2y·y' = 4 ⇒ y' = 2/y = 2/6 = 1/3 ✓.
```

7. Applications — Worked Exam Problems

Worked Example 10 — Area enclosed by a line and a parabola

Find the area enclosed between the parabola $y = x^2 - 2x - 3$ and the line $y = x - 3$.

```
1.  Find intersections:
2.     x² − 2x − 3 = x − 3  ⇒  x² − 3x = 0  ⇒  x = 0 or x = 3.
```

```
3.  On [0, 3] the line lies above the parabola (check at x = 1:
       line = −2, parabola = 1 − 2 − 3 = −4, so line > parabola). ✓
4.  Area = ∫₀³ [(x − 3) − (x² − 2x − 3)] dx
5.          = ∫₀³ [−x² + 3x] dx
6.          = [−x³/3 + 3x²/2]₀³
7.          = (−9 + 27/2) − 0
8.          = 9/2.

// The enclosed region is the "lens" between the secant chord and the parabola.
```

Worked Example 11 — Semicircle theorem in coordinates

Show that the triangle with vertices $A(-3,0), B(5,0), C(1,4)$ has a right angle at C, and deduce the equation of the circle through A, B, C.

```
1.  Vectors:  CA = A − C = (−4, −4);  CB = B − C = (4, −4).
2.  Dot product:  CA · CB = (−4)(4) + (−4)(−4) = −16 + 16 = 0.
3.  ∴ CA ⊥ CB, i.e. ∠ACB = 90°.
4.  By the converse of "angle in a semicircle", C lies on the
       circle with diameter AB.
5.  Centre = midpoint of AB = ((−3+5)/2, 0) = (1, 0).
6.  Radius = |AB|/2 = 8/2 = 4.
7.  Circle:  (x − 1)² + y² = 16.
8.  Check C(1,4):  0 + 16 = 16 ✓.
```

Worked Example 12 — Chord bisected by a perpendicular from centre

The circle $(x-3)^2 + (y-1)^2 = 25$ is cut by the line $y = 2x - 3$. Find the length of the chord and verify that the perpendicular from the centre to the chord bisects it.

```
1.  Substitute y = 2x − 3 into the circle equation:
2.     (x − 3)² + (2x − 4)² = 25
3.     (x² − 6x + 9) + (4x² − 16x + 16) = 25
4.     5x² − 22x + 0 = 0
5.     x(5x − 22) = 0  ⇒  x = 0 or x = 22/5.
```

```
Corresponding y:  y = −3  and  y = 29/5.
Chord endpoints:  P(0, −3), Q(22/5, 29/5).
Chord length:
   |PQ|² = (22/5)² + (29/5 + 3)²  = (22/5)² + (44/5)²
         = (484 + 1936)/25  =  2420/25  =  96.8.
   |PQ| = √96.8 = (22/5)√5.    // exact form
Midpoint M of PQ = (11/5, 7/5).
Vector from centre C(3, 1) to M:  (11/5 − 3, 7/5 − 1) = (−4/5, 2/5).
Direction of chord:  Q − P = (22/5, 44/5), parallel to (1, 2).
Dot product CM · (1, 2) = −4/5 + 4/5 = 0   ⇒   CM ⊥ chord ✓.
This confirms Theorem B: the perpendicular from the centre bisects the chord
```

8. Locus Problems

A *locus* is the set of points satisfying a geometric condition. Translating the condition into coordinates and simplifying usually produces a line or circle equation.

Worked Example 13 — Locus equidistant from two points

Find the locus of points $P(x, y)$ equidistant from $A(-2, 3)$ and $B(4, 1)$.

```
|PA|² = |PB|² translates the condition:
   (x + 2)² + (y − 3)² = (x − 4)² + (y − 1)²
   x² + 4x + 4 + y² − 6y + 9 = x² − 8x + 16 + y² − 2y + 1
   4x − 6y + 13 = −8x − 2y + 17
   12x − 4y − 4 = 0
   3x − y − 1 = 0.
∴ the locus is the straight line 3x − y − 1 = 0
    − this is the perpendicular bisector of AB.
Check: midpoint of AB = (1, 2) lies on 3(1) − 2 − 1 = 0 ✓.
    gradient of AB = (1 − 3)/(4 + 2) = −1/3;
    perpendicular gradient = 3 − matches line gradient ✓.
```

9. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Coordinate Geometry.

- **Lines:** know all three forms; convert to $ax + by + c = 0$ with integer coefficients when asked.
- **Gradient of perpendicular:** negative reciprocal. Check $m_1 m_2 = -1$ at the end.
- **Circle from general form:** complete the square in x and y separately; do not forget to subtract what you added.
- **Point inside/outside circle:** compare $(x_0 - a)^2 + (y_0 - b)^2$ with r^2.
- **Tangent at a point on a circle:** gradient is negative reciprocal of radius gradient; use point–slope.
- **Tangent from an external point:** set discriminant of line-into-circle quadratic to zero.
- **Line–circle intersection:** substitute y from the line into the circle, solve the resulting quadratic; sign of discriminant tells the geometry.
- **Parametric to Cartesian:** eliminate t by substitution or trig identity; state any range restriction on x, y.
- **Parametric $\mathrm{d}y/\mathrm{d}x$:** $(\mathrm{d}y/\mathrm{d}t)/(\mathrm{d}x/\mathrm{d}t)$; leave in terms of t.
- **Three points on a circle:** substitute into general form and solve for g, f, c, or use two perpendicular bisectors to find the centre.

Typical OCR examiner complaints (from mark reports).

- Writing a line equation in non-integer form when integer form was demanded.
- Arithmetic slips when completing the square for a circle — especially sign errors on the constant term.
- Using the "discriminant = 0" method for a tangent at a given *point on* the circle (over-complicated — use the perpendicular-radius method).
- Failing to state the parameter range, so the Cartesian equation describes more of the plane than the actual parametric curve.
- Dividing $\mathrm{d}y/\mathrm{d}t$ by $\mathrm{d}x/\mathrm{d}t$ in the wrong order, giving $\mathrm{d}x/\mathrm{d}y$ instead.
- Forgetting to verify that a point claimed to lie on a circle actually satisfies the equation before computing the tangent there.

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- Mishandling the external-tangent problem by assuming the tangent has a finite gradient, missing a vertical tangent case.

Formula snapshot (memorise).

- Gradient: $m = (y_2 - y_1)/(x_2 - x_1)$.
- Distance: $|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Midpoint: $M = ((x_1 + x_2)/2,\ (y_1 + y_2)/2)$.
- Perpendicular: $m_1 m_2 = -1$.
- Circle standard: $(x - a)^2 + (y - b)^2 = r^2$.
- Circle general: $x^2 + y^2 + 2gx + 2fy + c = 0$; centre $(-g, -f)$, $r^2 = g^2 + f^2 - c$.
- Parametric derivative: $\mathrm{d}y/\mathrm{d}x = (\mathrm{d}y/\mathrm{d}t)/(\mathrm{d}x/\mathrm{d}t)$.